Fiscal Year:	FY 2013	**Task Last Updated:**	FY 04/02/2013
PI Name:	O'Brian, Thomas Ph.D.		
Project Title:	NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boulder	**State:**	CO
Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/01/2012	**End Date:**	11/15/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:	NOTE: End date changed in November 2015 due to change in PI, per PI and new PI; original end date was 5/31/2017 (Ed., 12/17/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Jefferts, Steven (National Institute of Standards & Technology)		
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Performance Goal No.:			
Performance Goal Text:			
	To support primary ACES mission objectives, NIST proposes to: 1. Develop a Microwave Link Ground Terminal (MWL GT) at NIST in Boulder, CO to perform 10-17 uncertainty time and frequency comparisons between ACES clocks and world-leading optical and atomic clocks at NIST Boulder and the nearby JILA; and to support general ACES science goals. 2. Participate as fully as possible in coordinated ESA data analysis with formal ESA Data Analysis Centers and other ACES partners throughout the world. NIST will collaborate closely with ESA partners to define how NIST can best contribute to overall ACES mission goals within the constraints of NIST resources.		

Task Description:	As part of participation in the time and frequency transfer and data analysis, NIST will participate in tests of fundamental physics based on ACES data to include at least the following: • Global time and frequency transfer with uncertainty as low as 10-17. • Direct comparison of PHARAO space clock to NIST time and frequency standards at uncertainty of 10-17 or better. • Test of gravitation redshift at 2 x 10-6 uncertainty or better. • Search for variation in fundamental constants, such as fine structure constant at level of 10-18 per year or better. All these goals will be facilitated by the strong time and frequency infrastructure at NIST, including (a partial list): • World-class cesium microwave primary frequency standards, including NIST-F1 with fractional frequency uncertainty of 3 x 10-16, and NIST-F2 which will be operational in 2013 with an eventual uncertainty of 1 x 10-16 or better. • Two mercury ion optical clocks with current fractional frequency uncertainty of approximately 1 x 10-17 and continually improving. • Two aluminum ion “logic” optical clocks with current fractional frequency uncertainty of approximately 8 x 10-18 and continually improving. • Two ytterbium lattice optical clocks with uncertainties better than 1 x 10-16 and rapidly improving. • Two strontium lattice optical clocks with uncertainties approximately 3 x 10-17 and rapidly improving. (At JILA. A dedicated fiber optic system connects NIST and JILA to enable the JILA clocks to directly participate in NIST time and frequency transfer and intercomparisons.) • A high performance calcium optical frequency standard with exquisite short-term stability to serve as a “fly wheel” for clock comparisons. • A world-class time scale comprising seven hydrogen masers, with eight more masers to be added over the next three years. NIST scientists will participate in ESA/NASA planning meetings about the ACES program in general, and will specifically consult with ESA to determine how NIST can most fully participate in MLW GT data analysis and applications. In addition to the NIST resources already proposed for this program (NIST and JILA optical and microwave frequency standards, NIST time scale, NIST time and frequency transfer capabilities, etc.) NIST will directly contribute substantial funds and expert NIST staff time significantly exceeding the direct NASA investment. We propose the following revised program plan outline: Year 1 (2012): Begin design and construction of platform and other infrastructure for MWL hardware. $150k for partial payment of platform and related infrastructure design and construction. Year 2 (2013): Complete construction of platform and other infrastructure for MWL system. $150k for partial payment of platform and related infrastructure. Year 3 (2014): Receive MWL hardware. Install, test, and integrate MWL hardware into NIST time and frequency transfer system. Prepare and test MWL system for ACES time and frequency transfer. New NIST staff to prepare for participation in MWL data analysis. $150k for partial support of staff time. Year 4 (2015): ACES launch. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time. Year 5 (2016): On-going ACES experiments. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The major benefits and impacts of this research program can be grouped into three main areas: 1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements. 2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory. 3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, such search for possible Lorentz transformation variations and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosmology.

Task Progress:

Work on this project commenced in late May 2012. Most of the work on this task over this first year has focused on beginning coordination with international partners on long-term ACES project goals, and on initial development of the physical and human resources needed for NIST to serve as a host for a Microwave Link Ground Terminal (MWL GT) for precision time and frequency transfer, and to participate in data analysis needed for accurate and meaningful international time and frequency comparisons mediated by ACES at the 10-17 level – a factor of 10 or greater improvement over the current best international time and frequency transfer.
1. International program coordination.

• A kickoff meeting was held at NIST in Boulder, Colorado on June 29, 2012 to review plans for the overall ACES program and NIST's contributions, and to begin the international program coordination activities. Participants included representatives from NASA's Life and Physical Sciences Division and International Space Station research programs, representatives from the European Space Agency (in person and by teleconference), and NIST scientist participating in this research program and related programs. Participants shared information about NIST capabilities related to support ACES-mediated time and frequency transfer and research. These resources include (a partial list):

• Multiple world-leading frequency standards including two single-aluminum ion optical standards, two single-mercury ion optical standards, two ytterbium lattice optical standards, two strontium lattice optical standards, two cesium microwave primary standards, and multiple other frequency standards under development including calcium beam optical standards. The optical standards lead the world with frequency uncertainties in the low 10-17 range and better – the aluminum ion standards perform in the high 10-18 range. The microwave primary frequency standards are among the world leaders in the low 10-16 uncertainty range. Both the optical and microwave standards continue to improve and will likely be performing even better by the expected 2015 flight of the ACES project.

• A world-class time scale comprising multiple hydrogen masers to serve as a flywheel for comparison of frequency standards in the laboratory or through the MWL GT. NIST is in the midst of a 5 year, $5 million program to substantially upgrade the time scale and supporting infrastructure. The enhanced time scale should be operational by the expected 2015 flight of the ACES project, providing additional NIST support to ACES program goals.

• Advanced femtosecond laser frequency comb systems and technologies to enable the precise intercomparison of NIST optical frequency standards operating at very different frequencies, to permit the direct intercomparison of optical and microwave frequency standards, and to convert the optical frequency signals to the microwave range for accurate transmission through the MWL GT for international time and frequency coordination.

• Long and successful experience in microwave-based international time and frequency comparison and coordination through telecommunications satellites and Global Navigation Satellite Systems (GNSS, such as GPS, GLONASS, Galileo, etc.). NIST expertise in these areas will help ensure success of MWL GT operations and ACES-mediated international time and frequency transfer.

• Strong theory support for details of space-based time and frequency transfer, including leading expertise on relativity.

• Note that NASA and ESA are leveraging an enormous NIST investment in physical and human assets from a modest NASA investment to ensure the success of this project. The NIST existing investment supporting this project is orders of magnitude greater than the NASA investment in NIST.

• Co-PI Dr. Steven Jefferts presented the talk "International Time and Frequency Inter-Comparisons using ACES and the ISS" at the 1st Annual ISS Research and Development Conference in Denver, June 26-27, 2012. This talk outlined for the international ISS research community the NIST plans for the ACES project.

• Co-PI Dr. Steven Jefferts participated in the ISS ACES Mission International Working Group (IWG) meeting at SYRTE/Paris Observatory in Paris, December 4-5, 2012. The IWG meeting is a primary vehicle for international coordination of ACES/ESA/NASA project goals. Dr. Jefferts described NIST plans and participated in discussions on future international coordination.

• NIST project scientists regularly informally coordinate with international colleagues in the United States, Europe, and Asia on ACES project goals, sharing research progress, sharing information about project details, and sharing future plans.

2. Preparations for Microwave Link Ground Terminal (MWL GT) in Boulder, Colorado.

• The Microwave Link Ground Terminal (MWL GT) system will be provided by ESA to participating ground station partners across the world, including NIST in Boulder, Colorado. The MWL GT systems are still under development through ESA sponsorship. NIST has been closely coordinating with ESA contacts on details of the MWL GT hardware, information technology and infrastructure requirements.

• Siting the MWL GT at NIST. NIST scientists have been working with NIST facilities staff and engineering firms to develop the best location for the MWL GT that provides the best available views of the ISS as it passes by in orbit, and to develop the physical and support infrastructure needed to site the MWL GT on the roof of a NIST building, provide required utilities and information technology support, and provide other required infrastructure. After studying several alternative locations with different advantages and disadvantages, NIST has identified a location on the roof of the new NIST Precision Measurement Laboratory (PML) building. The $100 million PML was completed in 2012 and provides new laboratories with tight control of temperature, vibration, humidity and air quality. Most of the NIST optical and microwave frequency standards programs will be located in PML, as will most of the NIST time scale and time and frequency transfer operations. NIST and engineering firms are developing the final plans for the support and infrastructure needed for the MWL GT.

3. Preparations to support international time and frequency transfer and research through ACES.

• NIST project scientists have been closely coordinating with international colleagues on the plans for data exchange and analysis in the ACES project (mediated through the MWL GT), and how to best use and analyze that data for optimal international time and frequency transfer, and for research on relativity and other fundamental physics using ACES data.

• NIST scientists been analyzing data from the SIGI 2 GPS receiver on board the ISS, March 21-22, 2010, provided by

	Johnson Space Flight Center. NIST scientists are particularly analyzing how the data support precision determination of altitude of the ISS and thus ACES, since altitude differences as small as one meter represent a frequency shift of almost 1 x 10-16, about ten times greater than the desired time and frequency transfer results with ACES. NIST scientists Dr. Neil Ashby (also University of Colorado) and Dr. Marc Weiss will discuss these results at the 2nd Annual ISS Research and Development conference to be held in Denver July 16-18, 2013: “Testing the GR Redshift from the ISS.” This talk will support further discussions and analysis by the international ACES community on optimizing ACES-mediate time and frequency transfer and fundamental research. • NIST continues to perform research on and improve the performance of its several optical and microwave frequency standards that will be crucial to international time and frequency comparisons and ACES-mediated fundamental research: two single-aluminum ion optical standards, two single-mercury ion optical standards, two ytterbium lattice optical standards, two strontium lattice optical standards, two cesium microwave primary standards, and multiple other frequency standards under development including calcium beam optical standards. The optical standards lead the world with frequency uncertainties in the low 10-17 range and better – the aluminum ion standards perform in the high 10-18 range. The microwave primary frequency standards are among the world leaders in the low 10-16 uncertainty range. Both the optical and microwave standards continue to improve and will likely be performing even better by the expected 2015 flight of the ACES project. Eleven refereed papers were published about improvements in these frequency standards over the past year – please see the bibliography. • NIST continues to improve and modernize its time scale. NIST is on track to complete its major $5 million upgrade of the NIST time scale by the expected 2015 ACES launch. The improved time scale will further enhance NIST’s ability to support improved time and frequency transfer and research through ACES.
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